Notes: Bhandari, Kass, May, McGrattan & Schulz (JPE, 2026)

On the Nature of Entrepreneurship

Contents

1 Big picture and incremental contribution

- **Question.** What is the nature of entrepreneurship when it is measured using population-scale administrative tax data rather than household surveys?
- **Core empirical object.** The paper studies pass-through business owners: sole proprietors, partners, and S-corporation owners. These businesses account for a large share of U.S. private business activity but are often poorly measured in household surveys.
- **Main contrast with prior evidence.** Survey-based work often portrays the typical self-employed person as choosing entrepreneurship despite lower pay, high risk, liquidity constraints, and nonpecuniary motives. The IRS data imply a different picture when all dollars earned by the self-employed are counted.
- **Main findings.** The self-employed have higher average incomes, steeper life-cycle income profiles, and much more right-tail income than survey data suggest. They also face aggregate income risk, but the paper finds limited evidence that liquidity constraints, uninsurable risk, or nonpecuniary motives are first-order drivers of entry.
- **Dynamics.** Entry and exit rates in self-employment are stable over time, including during the Great Recession. Entrepreneurship is risky in income growth, but exits do not surge during downturns.
- **Pre-entry evidence.** Future entrepreneurs tend to have higher paid-employment income before entry than otherwise similar nonentrants. This weakens the common story that entry is mainly driven by job loss or inability to obtain paid work.
- **Policy relevance.** Conclusions about entrepreneurship depend heavily on whether one studies the typical person, the typical dollar, or the upper tail of entrepreneurial income.

1.1 Incremental contribution of the paper

- **Administrative data contribution.** The paper builds a longitudinal population-scale database of U.S. pass-through business owners from IRS data, overcoming small samples, short panels, and income top-coding in surveys.
- **Measurement contribution.** It clarifies how different tax-form definitions of self-employment change counts, incomes, and dynamics. The paper separately tracks self-employed income, paid-employed income, gross profits, and occupational information.
- **Distributional contribution.** It shows that survey samples miss much of the right tail of entrepreneurial income and understate losses in the left tail. This changes the interpretation of average returns and risk.
- **Life-cycle contribution.** It estimates age and time effects in levels for several employment groups, allowing income growth profiles for entrepreneurs to be separated from business-cycle shocks.
- **Risk contribution.** It distinguishes idiosyncratic income growth risk from aggregate downturn risk, and it uses the Great Recession to show that self-employed income is cyclically sensitive without producing large exit responses.
- **Entry contribution.** It uses pre-entry tax histories to show that entrants into self-employment tend to have higher paid-employment income before entry than later entrants, challenging the view that entrepreneurship is usually a fallback option.
- **Conceptual contribution.** It reframes the entrepreneurship literature away from the "typical self-employed individual" and toward the population of owners and dollars that account for business income.

2 Data and measurement

2.1 IRS administrative data and sample construction

The paper uses U.S. administrative tax data from the Internal Revenue Service. The core population is individuals aged 25–65, born between 1950 and 1975, alive in 2015, and observed in tax records. The authors then restrict to people with occupational information to create the main sample used for longitudinal analyses.

- The broad age 25–65 sample contains about 3.2 billion person-year observations.
- The age/cohort/alive-in-2015 restriction leaves roughly 2.0 billion observations.
- The main sample with occupational information contains about 1.3 billion person-year observations.
- Depending on the definition, the self-employed sample ranges from 97.7 million to 169.2 million person-year observations.
- The preferred main self-employment definition includes individuals satisfying any of three criteria: self-employed income, self-employed employee classification, or self-employment-related tax forms.

Interpretation: The data scale is the paper's central strength. It allows the authors to study rare high-income entrepreneurs, repeated entry and exit, industry detail, and long income histories that household surveys cannot capture reliably.

2.2 Self-employment definitions

The paper uses several definitions of self-employment:

- **Self-employed income criterion:** the person earns enough income from sole proprietorship, partnership, or S-corporation activity.
- **Self-employed employee criterion:** the person is paid by a pass-through business in which ownership is relevant.
- **Tax-form criterion:** the person files forms associated with pass-through business ownership.
- **Main definition:** an individual is self-employed if any of the three criteria is met.

Interpretation: This multiplicity of definitions is not just a data appendix issue. The measurement of entrepreneurship changes the apparent level of income, the distribution of income, and the comparison with survey samples.

2.3 Income variables

The main income variable is total income, built from tax records. The paper distinguishes:

- self-employed income;
- paid-employed income;
- gross profits;
- spousal wages;
- asset income;
- unemployment insurance income;
- adjusted gross income and other sources used in robustness and entry tests.

Because business income can be negative, the paper works with income levels and income changes rather than log income. This is especially important because entrepreneurial losses are common and large in the IRS data.

Interpretation: The level specification avoids log-transform distortions around zero and negative income. This is important for measuring both risk and life-cycle growth among business owners.

2.4 CPS comparison sample

The paper compares IRS self-employed samples with Current Population Survey samples. The CPS is a conventional source for studying self-employment, but it has smaller samples, top-coding, and limited measurement of business losses and pass-through income.

Interpretation: The CPS comparison is not merely descriptive. It explains why much prior evidence emphasized low returns and nonpecuniary motives: surveys overrepresent the center of the self-employed distribution and miss much of the right tail.

2.5 Balanced panel for life-cycle analysis

For life-cycle income profiles, the paper uses an IRS balanced panel with repeated observations. Individuals are grouped by employment status and characteristics. The main groups include:

- tried self-employment;
- primarily self-employed;
- mostly switching between self-employment and paid employment;
- primarily paid employed;
- not primarily employed.

Interpretation: The balanced panel lets the authors distinguish permanent individual differences, age growth, cohort structure, and time shocks. It is essential for separating entrepreneurial income growth from business-cycle variation.

3 Model and empirical specifications

3.1 Age-time income decomposition

Let i index individuals, t years, $a(i,t)$ age, $g(i)$ group, and $c(i)$ birth cohort. The paper specifies income in levels as

$$y_{it} = \alpha_i + b_{g(i),t} + \sum_{a=a_0}^{a(i,t)} \gamma^a_{c(i),g(i)} + \varepsilon_{it}. \tag{3}$$

The components are:

- α_i: individual fixed effect, including permanent ability, preferences, family inputs, and cohort level differences;
- $b_{g,t}$: group-specific time effect, including business-cycle shocks;
- $\gamma^a_{c,g}$: age effect for group g and cohort bin c;
- ε_{it}: residual income shock.

Interpretation: This is a rich decomposition for administrative panel data. It avoids log income, absorbs individual fixed effects, and allows entrepreneurial age effects and time effects to differ from those of paid workers.

3.2 First-differenced estimation

The first-differenced version is

$$\Delta y_{it} = \Delta b_{g(i),t} + \bar{\gamma}_{g(i)}^{a(i,t)} + \Delta \varepsilon_{it}. \tag{4}$$

The least-squares problem is

$$\min_{\{\Delta b_{g,t}, \bar{\gamma}_g^a\}} \sum_{g \in G} \sum_{t \in T} \sum_{i \in I} \left(\Delta y_{it} - \Delta b_{g(i),t} - \bar{\gamma}_{g(i)}^{a(i,t)} \right)^2 . \tag{5}$$

Interpretation: Differencing removes individual fixed effects. Identification of age effects requires overlapping cohorts and the assumption that age effects are the same across cohort bins of size at least two.

3.3 Integrated income profile

The estimated age and time effects are combined into integrated income profiles. For an aggregate group G, the profile starts from average income at age 25 and adds estimated growth rates:

$$Y_G(a) = Y_G(25) + \sum_{j=26}^{a} \sum_{g \in G} \frac{N_g^j}{\sum_k N_k^j} \left(\bar{\gamma}_g^j + \Delta b_{g,j} \right) . \tag{9}$$

Interpretation: The integrated profile is the paper's life-cycle object. It tells how the average income of a group evolves with age after accounting for time effects.

3.4 Income risk and implied risk aversion

The paper uses percentiles of income growth to assess whether entrepreneurial income risk can justify lower average earnings through compensating nonpecuniary benefits. A simple consumption-risk calculation links observed downside/upside risks to the degree of risk aversion needed to rationalize self-employment.

Interpretation: The risk analysis is deliberately tractable. The goal is not to estimate a full structural model, but to see whether income risk alone can rationalize survey-style claims that entrepreneurs accept lower pay for nonpecuniary benefits.

3.5 Entry and exit dynamics

The paper estimates self-employment exit rates and entry rates over time using the IRS panel. It also compares current entrants with future entrants using past income histories.

Interpretation: Stable entry and exit through the Great Recession is important because it weakens the view that entrepreneurship mainly absorbs unemployed workers during downturns.

4 Identification strategy

4.1 Measurement identification through administrative data

The first identification issue is not causal treatment assignment but measurement. The paper identifies the distribution and dynamics of entrepreneurship by observing the full tax population rather than a survey sample.

- IRS data measure the high-income right tail better than surveys.
- IRS data capture large negative business income that surveys underrepresent.
- IRS data allow consistent definitions of self-employment based on business income and tax forms.

Interpretation: The paper's key identification contribution is to show that a different data-generating system changes the facts. Many prior conclusions are sensitive to survey measurement.

4.2 Age and time effects

The age-time decomposition identifies life-cycle growth by using multiple cohorts observed in overlapping years. The key assumption is that age effects are common within cohort bins, while cohort level differences are absorbed by individual fixed effects.

Interpretation: This strategy is not a causal estimate of entrepreneurship on income. It is a decomposition that asks how incomes evolve for observed self-employed and paid-employed groups, after removing individual fixed effects and group-specific time effects.

4.3 Risk and recession evidence

Time effects identify aggregate income risk by group. The Great Recession is especially informative because it occurs in the middle of the panel and creates a large aggregate shock.

Interpretation: The recession evidence shows that self-employed incomes fall much more than paid-employed incomes, but self-employment exit rates remain stable. This separates income risk from exit behavior.

4.4 Entry selection

The comparison of current entrants with future entrants uses pre-entry paid income. If entrepreneurship were primarily a fallback after poor labor-market performance, entrants would be expected to have lower past paid income than similar nonentrants.

Interpretation: The opposite pattern supports positive selection into entrepreneurship, at least among entrants who can be tracked in IRS data.

4.5 Remaining limitations

- Tax data do not include employer benefits and may still contain income underreporting.
- Pass-through owners are not all entrepreneurs in the Schumpeterian sense; some are professional practices or closely held businesses.

- The estimates are descriptive/decompositional rather than experimental causal effects of becoming self-employed.
- Nonpecuniary motives may matter for some marginal entrants even if they do not explain aggregate dollars earned in self-employment.

Interpretation: These limits are important. The paper changes the empirical facts about entrepreneurship, but it does not claim that every business owner is a high-growth entrepreneur or that all nonpecuniary motives are irrelevant.

5 Tables and figures

5.1 Table 1: IRS Sample Selection and Table 2: Summary Statistics for Main IRS Sample

Read:

- Table 1 shows the stepwise construction from all age-eligible individuals to the main sample with occupational information and self-employment definitions.
- Table 2 shows that self-employed individuals represent 8.4% of counts but 15.2% of total income and 97.4% of self-employed income.
- Mean total income is about $88,600 for the self-employed versus $56,400 for paid-employed individuals.
- The right tail is striking: the self-employed 90th percentile of total income is about $193,300, almost twice that of paid-employed individuals.
- The self-employed are more male, more often married, more likely to have children, and much more likely to have employees and gross profits.

Interpretation: Table 1 is a measurement map; Table 2 is the first major substantive fact. Self-employed people are a small share of counts but a large share of income, especially in the upper tail.

Economic message: The administrative-data picture differs sharply from survey narratives: self-employment is not mostly low-income subsistence work when weighted by dollars.

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TABLE 1
IRS SAMPLE SELECTION

Sample, Description	Observations
	A. All Individuals
A. Aged 25–65	3.2 billion
B. In sample A, born in years 1950–75, and alive in 2015	2.0 billion
C. In sample B and has occupational information	1.3 billion
	B. Self-Employed Individuals ($\lvert y^{SE} \rvert > 5{,}000$)
1. In sample A and self-employed income criteria met	169.2 million
2. In sample C and self-employed income criteria met	97.7 million
3. In sample C and any of three self-employed criteria met	107.6 million
All three self-employed criteria met	29.3 million
Self-employed income and gross profit criteria met	64.6 million
All other cases	13.7 million

NOTE.—Data for sample A include all living individuals aged 25–65 with a US Social Security number over the sample period 2000–2015. Samples B and C are balanced panels. Income from self-employment is denoted y^{SE}. For individuals to be included in one of the self-employed samples, the absolute value of the self-employment income, $\lvert y^{SE} \rvert$, must exceed US$5,000 in 2012 dollars. Three additional criteria are checked. If $\lvert y^{SE} \rvert$ is greater than income from paid employment, then the self-employed income criteria is met. If gross profits

(a) Table 1: IRS Sample Selection

TABLE 2
SUMMARY STATISTICS FOR MAIN IRS SAMPLE

	Total Sample (1)	Self-Employed (2)	Paid Employed (3)	Nonemployed (4)
Observations (millions)	1,279.9	107.6	940.9	231.4
Shares (%):				
Counts	100	8.4	73.5	18.1
Total income	100	15.2	84.6	.2
Self-employed income	100	97.4	2.4	.2
Paid-employed income	100	2.7	97.1	.2
Income (2012 US$1,000s):				
Mean total income	49.0	88.6	56.4	.6
10th percentile	.0	5.9	14.0	.0
25th percentile	11.4	12.7	24.8	.0
50th percentile	32.8	29.0	41.4	.0
75th percentile	58.6	78.4	65.3	.3
90th percentile	95.3	193.3	100.1	3.0
Mean self-employed income	6.5	75.0	.2	.1
10th percentile	.0	−6.0	.0	.0
25th percentile	.0	10.2	.0	.0
50th percentile	.0	23.1	.0	.0
75th percentile	.0	64.1	.0	.0
90th percentile	2.9	167.8	.0	.0
Mean paid-employed income	42.6	13.6	56.2	.5
10th percentile	.0	.0	14.0	.0
25th percentile	5.2	.0	24.8	.0
50th percentile	30.1	.0	41.4	.0
75th percentile	55.3	1.4	65.2	.0
90th percentile	88.2	29.2	99.6	2.6
Education and skills (%):				
College educated	52.8	56.5	56.4	36.7
Cognitive	52.4	59.0	55.3	37.8
Interpersonal	58.6	56.8	62.5	43.7
Manual	37.6	39.2	37.1	39.1
Primary industry (%):				
Agriculture	.7	1.4	.8	
Mining	.3	.4	.3	
Utilities	.1	.1	.2	
Construction	4.7	15.1	4.7	
Manufacturing	7.7	2.8	10.2	
Wholesale trade	2.5	2.6	3.1	
Retail trade	5.3	7.5	6.4	
Transportation	2.1	5.5	2.3	
Information	1.0	1.1	1.3	
Finance	2.1	3.2	2.5	
Real estate	1.5	4.9	1.5	
Professional services	6.0	13.4	6.6	
Management	.4	.1	.6	
Administration	2.9	4.9	3.4	
Education	.3	.6	.3	
Health care	4.0	8.3	4.5	
Arts	.6	2.1	.6	
Accommodation	2.4	3.7	2.9	
Other services	2.4	11.5	1.9	
Other NAICS code	9.1	3.0	12.1	
Missing NAICS code	43.7	7.8	34.0	100

(b) Table 2: Summary Statistics, part 1

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TABLE 2 (*Continued*)

	Total Sample (1)	Self-Employed (2)	Paid Employed (3)	Nonemployed (4)
Employees and profits:				
Has employees (%)	3.2	32.9	.4	.7
Gross profits (2012 US$1,000s)	22.2	249.0	1.1	2.5
Demographics:				
Male (%)	50.2	73.0	50.8	37.0
Married (%)	61.2	67.2	62.5	53.0
Has children (%)	50.6	58.9	53.2	36.1
Mean number of children	1.0	1.1	1.0	.7
Median birth year	1963	1962	1963	1964
Other income (2012 US$1,000s):				
Mean spousal wages	26.7	21.7	25.2	35.0
Mean asset income	8.8	37.5	5.2	9.8
Mean unemployment insurance income	.4	.2	.4	.6

NOTE.—The summary statistics are computed using individual-year observations in the IRS sample C in table 1, with the criteria for self-employed in (3). See sec. II for more details on the sample construction and income measures. Incomes and gross profits are reported in thousands of 2012 dollars. To ensure that no confidential information is disclosed, reported percentiles here and elsewhere are computed as an average of observations around the value listed in the table. Industries are classified by two-digit NAICS code.

In terms of demographics, we find that a larger share of the self-employed population are male and married with children (listed as non-spousal dependents aged 25 or younger). Finally, in terms of other incomes, the self-employed file tax returns with lower average spousal incomes and higher average asset incomes compared with other groups.

III. Comparison to Current Population Survey

In this section, we compare cross-sectional moments for the IRS samples described in section II with survey data from the CPS and show that there

(a) Table 2 continued.

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TABLE 3
CPS AND IRS SELF-EMPLOYED SAMPLE COMPARISON

	CPS Sample		IRS Sample		
	1 (1)	2 (2)	1 (3)	2 (4)	3 (5)
Observations (millions)	85.1	138.3	169.2	97.7	107.6
Income (2012 US$1,000s):					
Mean total income	49.6	66.2	74.1	85.4	88.6
10th percentile	8.5	10.6	.0	5.5	5.9
25th percentile	15.7	20.4	10.5	11.8	12.7
50th percentile	30.1	38.9	22.0	26.4	29.0
75th percentile	54.6	74.0	60.7	73.9	78.4
90th percentile	100.4	138.1	159.9	190.3	193.3
Mean self-employed income	47.6	65.2	70.3	80.9	75.0
10th percentile	8.4	10.4	−5.0	5.4	−6.0
25th percentile	15.3	20.2	10.0	11.0	10.2
50th percentile	29.6	38.8	21.0	25.1	23.1
75th percentile	53.2	72.9	57.7	69.8	64.1
90th percentile	98.2	135.8	152.0	180.8	167.8
College educated (%)	40.7	46.8	NA	54.8	56.5
Top NAICS codes:					
First	23	23	23	23	23
Second	81	44	54	54	54
Third	44	54	81	81	81
Fourth	56	81	62	44	44
Fifth	62	62	44	62	62
Demographics:					
Male (%)	64.9	68.0	72.6	72.7	73.0
Married (%)	70.4	74.4	62.5	66.0	67.2
Birth year	1961	1961	1961	1962	1962

NOTE.—Summary statistics are reported for two CPS samples that differ in the criteria used to assign individuals in a particular year as self-employed. CPS sample 1 uses business income reported in the survey and categorizes an individual as self-employed if the absolute value of the income is greater than US$5,000 (in 2012 dollars) and greater than income from paid employment. The same criteria are used for the self-employed for CPS sample 2 after reclassifying the wage and salary income of incorporated business owners as income from self-employment. The criteria used to assign individual-year observations in IRS self-employed sample 1 are the same as for CPS sample 1. Sample criteria for IRS samples 2 and 3 are the same as in table 1. To ensure that no confidential information is disclosed, reported IRS percentiles are computed as an average of observations around the value listed in the table. NA = not applicable.

(b) Table 3: CPS and IRS self-employed sample comparison.

5.2 Figure 1: Income by Age in CPS and IRS Samples and Tables 3–4: CPS/IRS Comparisons

Read:

- Figure 1 compares IRS and CPS income-by-age profiles for self-employed and paid-employed individuals.
- The IRS profiles show much higher self-employed incomes at older ages than the CPS profiles.
- Table 3 shows that CPS self-employed samples have much lower mean total income than comparable IRS samples.
- Table 4 shows that the CPS underweights the income share of the top self-employed tail and also misses large negative-income observations in the left tail.

Interpretation: Together, these visuals explain why prior survey-based conclusions differ. The CPS sees many self-employed individuals, but it misses or compresses the tails where much entrepreneurial income and risk are located.

Economic message: The self-employment distribution is tail-driven. For entrepreneurship, the typical person and the typical dollar tell different stories.

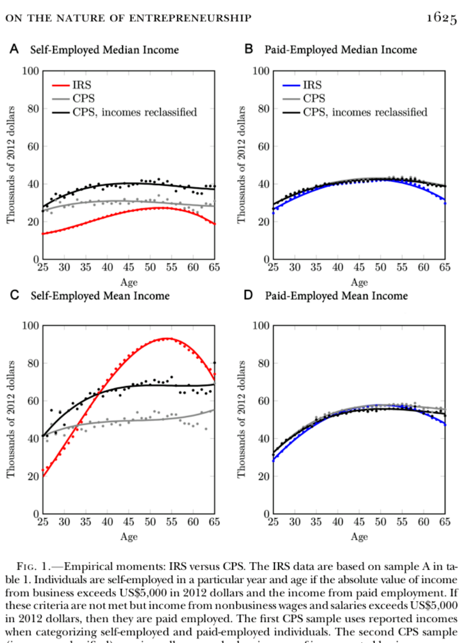

Fig. 1.—Empirical moments: IRS versus CPS. The IRS data are based on sample A in table 1. Individuals are self-employed in a particular year and age if the absolute value of income from business exceeds US$5,000 in 2012 dollars and the income from paid employment. If these criteria are not met but income from nonbusiness wages and salaries exceeds US$5,000 in 2012 dollars, then they are paid employed. The first CPS sample uses reported incomes when categorizing self-employed and paid-employed individuals. The second CPS sample

(a) Figure 1: Income by age in CPS and IRS samples.

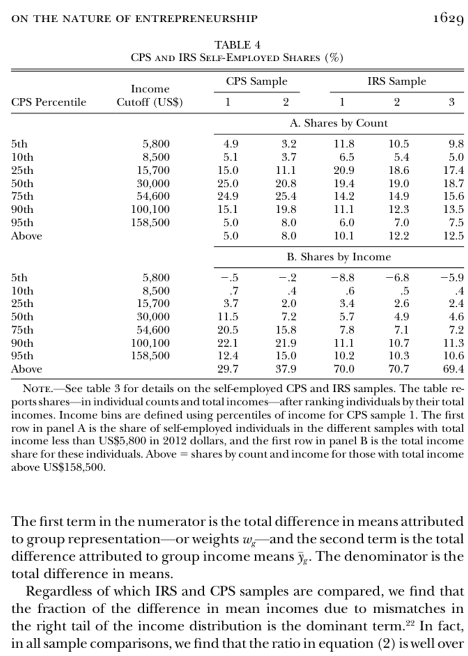

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TABLE 4
CPS and IRS Self-Employed Shares (%)

CPS Percentile	Income Cutoff (US$)	CPS Sample 1	CPS Sample 2	IRS Sample 1	IRS Sample 2	IRS Sample 3
		A. Shares by Count				
5th	5,800	4.9	3.2	11.8	10.5	9.8
10th	8,500	5.1	3.7	6.5	5.4	5.0
25th	15,700	15.0	11.1	20.9	18.6	17.4
50th	30,000	25.0	20.8	19.4	19.0	18.7
75th	54,600	24.9	25.4	14.2	14.9	15.6
90th	100,100	15.1	19.8	11.1	12.3	13.5
95th	158,500	5.0	8.0	6.0	7.0	7.5
Above		5.0	8.0	10.1	12.2	12.5
		B. Shares by Income				
5th	5,800	−.5	−.2	−8.8	−6.8	−5.9
10th	8,500	.7	.4	.6	.5	.4
25th	15,700	3.7	2.0	3.4	2.6	2.4
50th	30,000	11.5	7.2	5.7	4.9	4.6
75th	54,600	20.5	15.8	7.8	7.1	7.2
90th	100,100	22.1	21.9	11.1	10.7	11.3
95th	158,500	12.4	15.0	10.2	10.3	10.6
Above		29.7	37.9	70.0	70.7	69.4

Note.—See table 3 for details on the self-employed CPS and IRS samples. The table reports shares—in individual counts and total incomes—after ranking individuals by their total incomes. Income bins are defined using percentiles of income for CPS sample 1. The first row in panel A is the share of self-employed individuals in the different samples with total income less than US$5,800 in 2012 dollars, and the first row in panel B is the total income share for these individuals. Above = shares by count and income for those with total income above US$158,500.

The first term in the numerator is the total difference in means attributed to group representation—or weights w_g—and the second term is the total difference attributed to group income means $\bar{y}_g$. The denominator is the total difference in means.

Regardless of which IRS and CPS samples are compared, we find that the fraction of the difference in mean incomes due to mismatches in the right tail of the income distribution is the dominant term.[22] In fact, in all sample comparisons, we find that the ratio in equation (2) is well over

(b) Table 4: CPS and IRS self-employed shares.

5.3 Tables 5–6 and Figures 2–3: Age and Time Effects

Read:

- Table 5 reports age effects. Self-employed age effects are higher than paid-employed age effects at many prime working ages.
- Table 6 reports time effects. Self-employed income growth is much more volatile over time and falls sharply in 2008–9.

- Figure 2 integrates age and time effects into income profiles: by age 55, self-employed income is about \$134,000 versus about \$79,000 for paid-employed workers.
- Figure 3 shows that self-employed incomes experience much larger recession-year declines than paid-employed incomes.

Interpretation: The evidence reconciles high returns with high aggregate risk. Entrepreneurs have steeper life-cycle income growth, but they are also more exposed to macroeconomic downturns.

Economic message: The entrepreneurial income premium is not risk-free. The key fact is that higher average growth coexists with larger cyclical exposure.

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TABLE 5
ESTIMATES OF AGE EFFECTS (2012 US$)

Age	Self-Employed	Paid Emplc
26	5,242	3,211
27	5,405	2,957
28	5,189	2,712
29	5,000	2,498
30	4,990	2,286
31	4,552	2,185
32	4,910	2,081
33	5,142	1,858
34	5,060	1,858
35	4,603	1,689
36	4,218	1,536
37	3,939	1,412
38	3,750	1,281
39	3,743	1,228
40	3,538	1,171
41	3,220	1,024
42	2,542	921
43	2,862	735
44	1,417	602
45	1,744	461
46	744	292
47	894	194
48	685	70
49	775	−17
50	26	−316
51	148	−426
52	−666	−532
53	−1,000	−563
54	−1,165	−833
55	−1,432	−692
56	−2,042	−1,202
57	−3,443	−1,377
58	−2,786	−1,482
59	−3,517	−1,737
60	−4,632	−2,361
61	−4,154	−3,169
62	−6,410	−4,239
63	−6,510	−5,786
64	−7,449	−4,871
65	−8,145	−5,464

NOTE.—Age effects ($\bar{\gamma}_g^a$) are reported in 2012 dollars for two ag gated groups: individuals in the tried self-employment subgroup (employed) and individuals in the primarily paid-employed subgr (paid employed). See details of the group assignment in sec. IV.B the estimation procedure in sec. IV.A.

for the self-employed compared with paid-employed
nains higher until age 52. Second, the estimated time
employed are much more volatile and significantly lo
at Recession, which occurs midway through the samp

(a) Table 5: Estimates of age effects.

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TABLE 6
ESTIMATES OF TIME EFFECTS (2012 US$)

ar	Self-Employed	Paid Employec
'01	1,353	721
'02	−686	255
'03	7,341	895
'04	−775	1,322
'05	5,416	1,055
'06	3,857	1,575
'07	−1,528	1,758
'08	−9,655	−373
'09	−8,785	−1,583
'10	1,661	883
'11	2,245	667
'12	10,306	672
'13	−2,846	−188
'14	4,579	1,123
'15	3,939	1,416

NOTE.—Time effects ($\Delta\beta_{g,t}$) are reported in 2012 dollars for two aggre ted groups: individuals in the tried self-employment subgroup (self nployed) and individuals in the primarily paid-employed subgroup aid employed). See details of the group assignment in sec. IV.B anc e estimation procedure in sec. IV.A.

the estimates of the age and time effects to construct
rpretable integrated income profiles for different ag
interest. Let $\mathcal{G}$ be one of these aggregated groups. To
ated income profile for $\mathcal{G}$, we start by computing the av

(b) Table 6: Estimates of time effects.

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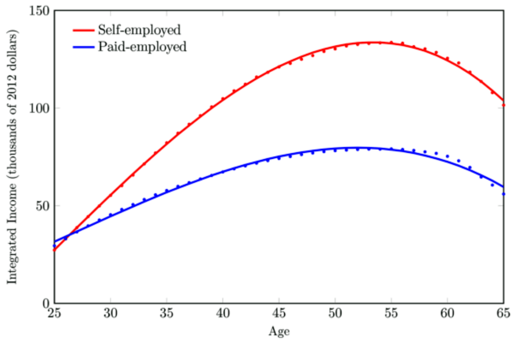


FIG. 2.—Estimated income profiles for self-employed and paid-employed individuals. The figure shows integrated income $Y\mathcal{G}(a)$ at each age a for two aggregated groups $\mathcal{G}$: individuals in the tried self-employment subgroup (self-employed) and individuals in the primarily paid-employed subgroup (paid employed). Integrated income is defined in equation (9). See details of the group assignment in section IV.B and the estimation procedure in section IV.A.

to account for business income underreporting. As we noted earlier, the BEA adjusts total business incomes in the US national accounts using net misreporting rates of 46% for unincorporated owners and 14% for

(a) Figure 2: Estimated income profiles.

The figure shows large differences in growth for the two groups during the 2008–9 downturn. Incomes were down: 10% of average income for self-employed individuals in 2008 and imperceptibly for paid-employed

(b) Figure 3: Estimated time effects relative to average income.

5.4 Table 7: Statistics after Grouping Individuals in IRS Balanced Panel

Read:

- Table 7 separates primarily self-employed individuals into those earning above and below paid-employed peers, mostly switching individuals, primarily paid employed, and not primarily employed.
- Primarily self-employed individuals earning above paid-employed peers are a small share of people but account for a large share of self-employed income.
- This group has very high mean total income, high gross profits, high employee incidence, and high college-educated and cognitive-skill shares.
- Those earning below paid-employed peers look very different, with lower income, fewer employees, and different industry composition.

Interpretation: Table 7 decomposes heterogeneity inside self-employment. It shows why statements about "the self-employed" are misleading unless one distinguishes high-income owners, low-income owners, and switchers.

Economic message: The upper tail of entrepreneurs looks selected on skills and scale, while other self-employed groups look closer to low-income or transitional work.

TABLE 7

Statistics after Grouping Individuals in IRS Balanced Panel

	Primarily Self-Employed			Mostly Switching Self-Employed/ Paid Employed (4)	Primarily Paid Employed (5)	Not Primarily Employed (6)
	Total (1)	Earning Above Paid Employed (2)	Earning Below Paid Employed (3)			
Individuals (millions)	2.8	1.2	1.6	3.4	47.2	26.6
Shares (%):						
Counts	3.4	1.5	1.9	4.3	59.0	33.3
Total income	9.4	7.9	1.5	7.3	72.2	11.1
Self-employed income	62.3	52.6	9.5	21.4	5.3	11.0
Paid-employed income	1.4	1.1	.3	5.1	82.3	11.1
Income (2012 US$1,000s):						
Mean total income	134.2	262.1	37.8	83.3	60.0	16.4
10th percentile	15.0	54.3	11.8	14.8	21.0	1.6
25th percentile	26.7	79.6	18.2	23.4	30.3	5.0
50th percentile	56.6	144.9	30.3	41.7	45.3	10.8
75th percentile	130.1	275.7	50.6	81.8	68.0	20.0
90th percentile	291.3	528.0	84.6	164.8	102.5	33.5
Mean self-employed income	117.0	230.4	31.6	32.4	.6	2.1
10th percentile	12.3	43.1	9.6	−.1	−.1	.0
25th percentile	22.1	66.4	15.2	4.5	.0	.0
50th percentile	47.5	124.0	25.8	11.3	.0	.0
75th percentile	112.3	242.8	43.8	29.7	.0	.6
90th percentile	256.9	469.0	74.0	73.2	1.0	4.6
Mean paid-employed income	17.2	31.8	6.3	50.9	59.4	14.2
10th percentile	.0	.0	.0	8.8	21.0	.6
25th percentile	.0	.0	.0	15.1	30.3	3.5
50th percentile	2.6	6.1	1.5	27.0	45.1	9.3
75th percentile	11.6	24.3	6.5	50.2	67.5	18.3
90th percentile	35.2	68.6	16.0	93.5	101.3	30.9

Table 7: Statistics after grouping individuals in IRS balanced panel, part 1.

TABLE 7 (*Continued*)

	Primarily Self-Employed			Mostly Switching Self-Employed/ Paid Employed (4)	Primarily Paid Employed (5)	Not Primarily Employed (6)
	Total (1)	Earning Above Paid Employed (2)	Earning Below Paid Employed (3)			
Education and skills (%):						
College educated	62.4	73.4	54.2	63.6	59.7	38.4
Cognitive	61.4	63.1	60.3	62.7	57.7	40.9
Interpersonal	60.5	71.3	52.5	63.4	65.6	45.5
Manual	36.0	29.4	40.9	36.9	36.1	40.6
Primary industry (%):						
Agriculture	1.8	2.0	1.6	1.2	.9	1.3
Mining	.5	.5	.4	.5	.5	.3
Utilities	.1	.0	.0	.1	.3	.1
Construction	17.9	15.1	20.2	13.5	5.6	8.3
Manufacturing	3.6	4.5	2.9	6.2	14.2	8.4
Wholesale trade	3.4	4.6	2.4	3.7	4.0	3.0
Retail trade	8.1	9.1	7.5	8.6	8.3	10.0
Transportation	5.1	2.8	7.0	5.8	3.1	3.3
Information	1.0	.9	1.0	1.6	1.8	1.3
Finance	4.2	3.7	4.6	3.9	3.7	2.5
Real estate	5.0	4.6	5.4	4.1	1.8	2.7
Professional services	15.5	18.8	13.0	15.5	8.9	7.8
Management	.1	.1	.1	.2	.9	.4
Administration	4.3	4.1	4.5	4.6	4.0	6.6

(a) Table 7 continued, part 2.

Education	.4	.4	.4	.6	.5	.6
Health care	9.5	13.3	6.7	8.6	5.6	7.1
Arts	1.9	1.5	2.3	1.7	.8	1.2
Accommodation	3.6	4.2	3.2	4.4	3.5	6.2
Other services	11.1	8.6	13.0	8.0	2.2	5.5
Other NAICS code	.7	.3	.6	5.0	17.1	10.1
Missing NAICS code	2.3	1.0	3.3	2.1	12.2	13.3
Employees and profits:						
Ever had employees (%)	63.9	82.4	50.0	47.9	4.7	8.2
Gross profits (2012 US$1,000s)	400.7	732.9	147.9	156.5	6.7	19.0
Demographics:						
Male (%)	81.6	79.5	83.4	72.9	52.5	39.9
Married (%)	78.0	81.8	75.4	70.3	66.8	56.1
Has children (%)	84.8	85.3	84.7	85.2	82.5	80.7
Mean number of children	2.2	2.2	2.3	2.4	2.2	2.3
Median birth year	1960	1960	1960	1964	1963	1964
Other income (2012 US$1,000s):						
Mean spousal wages	26.3	32.1	22.0	27.3	30.6	38.0
Mean asset income	56.0	90.8	29.6	21.4	4.6	9.5
Mean unemployment insurance income	.1	.1	.1	.4	.4	.5

Note.—The summary statistics are computed using panel data for individuals in IRS sample C in table 1, with the criteria for self-employed in (3). See sec. IV for more details on the sample construction and income measures. For more details on the time-invariant groups, see sec. IV.B. Incomes and gross profits are reported in thousands of 2012 dollars. To ensure that no confidential information is disclosed, reported percentiles are computed as an average of observations around the value listed in the table. Industries are classified by two-digit NAICS code.

(b) Table 7 continued, part 3.

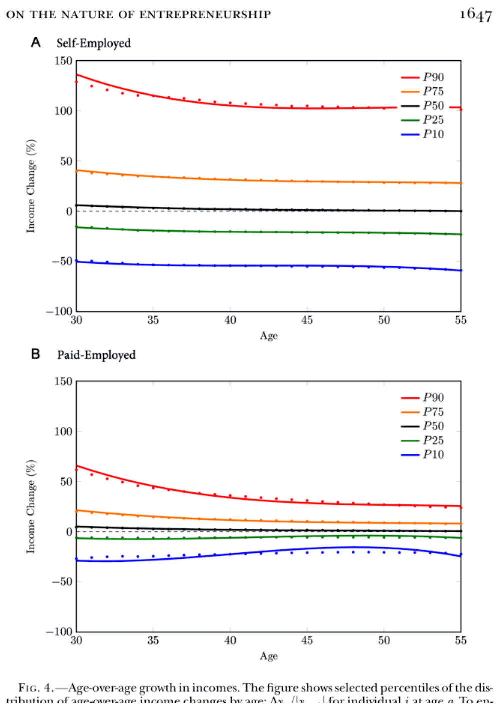
5.5 Figure 4 and Figure 5: Income Growth Risk and Implied Risk Aversion

Read:

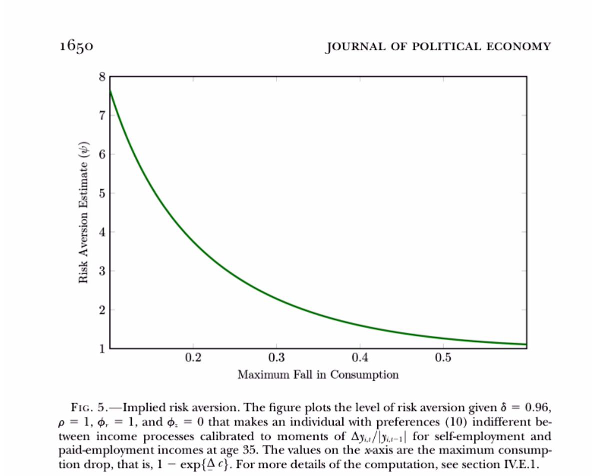
- Figure 4 plots percentiles of age-over-age income growth by age for self-employed and paid-employed workers.
- Growth dispersion is much wider for the self-employed, especially in the lower and upper tails.
- Figure 5 computes the implied risk aversion needed for income risk to rationalize entrepreneurship under different assumptions.
- Under plausible parameters, income risk alone does not make self-employment look unattractive enough to require large nonpecuniary benefits.

Interpretation: The figures directly engage the standard compensating-differentials story. Entrepreneurs face more risk, but the risk-return trade-off in the IRS data is not consistent with the claim that the typical dollar in entrepreneurship is earned at a lower financial return.

Economic message: The risk is real, but so is the return. Administrative data weaken the hypothesis that nonpecuniary benefits are the primary explanation for entrepreneurship.



(a) Figure 4: Age-over-age growth in incomes.



(b) Figure 5: Implied risk aversion.

5.6 Table 8 and Figures 6–7: Cyclical Sensitivity, Exit, and Entry

Read:

- Table 8 shows that during 2008–9, self-employed groups in cyclically sensitive sectors suffer much larger income declines than paid-employed groups.
- Figure 6 shows that self-employment exit rates remain broadly stable from 2001 to 2015, despite the Great Recession.
- Figure 7 shows self-employment shares and entry rates are also stable over the sample period.
- The self-employed income response to the Great Recession is large, but the extensive margin of self-employment does not collapse.

Interpretation: This combination is important: entrepreneurs bear substantial income risk without responding through massive exit. This suggests attachment to ongoing businesses, fixed costs, option value, or other frictions in exit.

Economic message: Entrepreneurship is cyclically risky, but entry and exit are not simply residual labor-market states that mechanically absorb unemployment shocks.

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TABLE 8
Estimated Time Effects Relative to Average Income (%), 2008–9 Cyclically Sensitive Groups by Education and Industry

	Primarily Self-Employed	Mostly Switching	Primarily Paid Employed
College educated:			
Real estate	−50	−18	−4
Construction	−33	−16	−4
Accommodation	−22	−12	−2
Manufacturing	−21	−14	−3
Retail trade	−14	−10	−3
Professional services	−7	−6	−1
Not college educated:			
Manufacturing	−24	−9	−3
Construction	−15	−9	−5
Agriculture	−7	0	−1
Retail trade	−6	−4	0
Transportation	−4	−4	−2
Other services	−2	−2	0

Note.—The table reports weighted averages of estimated time effects for select groups g at time t, i.e., $\Delta\beta_{g,t}$, which are divided by average income $\bar{y}_{g,t}$. Weights are constructed from group counts.

(a) Table 8: Time effects during 2008–9 for cyclically sensitive groups.

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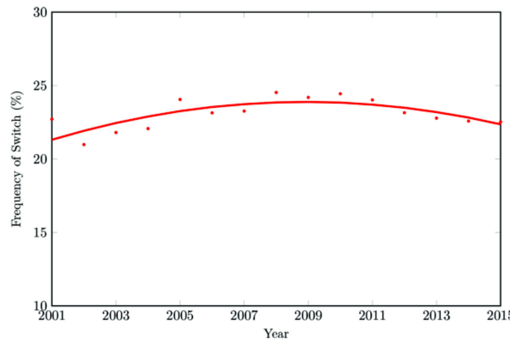

Fig. 6.—Self-employment exit rates, 2001–15. The figure shows the fraction of individuals whose status is self-employed in year $t-1$ and paid employed or nonemployed in year t. Age and time effects are extracted using the procedure outlined in section IV.A.

(b) Figure 6: Self-employment exit rates.

While the evidence on entry rates provides a prima facie case that there are few impediments to entrepreneurship, we turn next to more direct

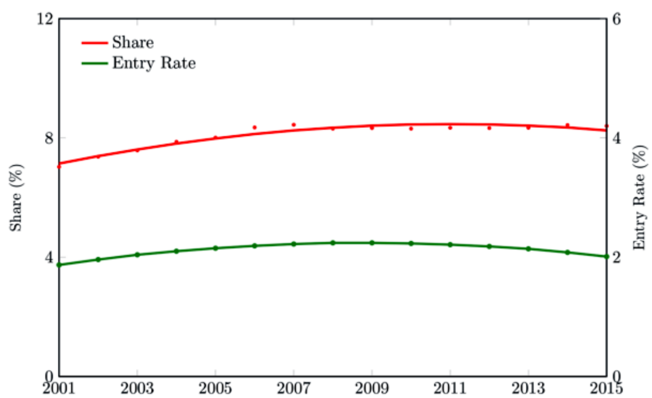

Figure 7: Self-employment shares and entry rates.

5.7 Tables 9–10: Past Incomes of Switchers and First Positive Income for Founders

Read:

- Table 9 compares current switchers with future switchers and shows that current entrants often have higher pre-entry paid income than people who enter later.
- This pattern is inconsistent with a story where most entrepreneurial entry comes from weak labor-market prospects or low outside options.
- Table 10 shows that most S-corporation and partnership founders have positive total income immediately, even if business net income itself takes longer to become positive.
- Positive owner income is often supported by wages, spousal income, and asset income during the early business years.

Interpretation: Table 9 is the paper's strongest evidence against the fallback-entry narrative. Table 10 shows why liquidity constraints may not bind as often as survey narratives imply: many founders have other income sources during the early years.

Economic message: Entrepreneurial entry looks positively selected in paid-employment income, and owners often have income buffers when business income is initially weak.

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TABLE 9
DIFFERENCES IN PAST INCOMES FOR SWITCHERS BY AGE, CURRENT VERSUS FUTURE SWITCHER (2012 US$1,000s)

Past Income Type, Percentile	Age 30	40	50	60
Interest income:				
25th	−.1	−.4	−1.1	−1.8
50th	.0	−.1	−.2	−.4
75th	.0	.0	.0	.0
Dividend income:				
25th	−.1	−.3	−.9	−1.6
50th	.0	.0	−.2	−.3
75th	.0	.0	.0	.0
Capital gains:				
25th	−.2	−.9	−3.6	−6.3
50th	.0	−.1	−.5	−1.1
75th	.0	.0	.0	.2
Spousal wages:				
25th	−4.7	−14.6	−19.5	−20.3
50th	−.9	−2.4	−2.6	−1.8
75th	.0	.0	.0	.0
Own wages:				
25th	−.3	.3	.3	1.5
50th	.0	1.0	1.1	2.4
75th	.3	2.5	3.4	5.1

NOTE.—The table reports the time-averaged past incomes (in thousands of 2012 dollars) of current switchers less the time-averaged past incomes of peers who have similar characteristics but switch later. The sample used to construct the results includes individuals with at most one observed switch between paid employment and self-employment. Differences in incomes are shown for selected percentile ranks and ages. To ensure that no confidential information is disclosed, reported percentiles are computed as an average of observations around the value listed in the table.

comparisons, we conclude that most switchers are negatively selected

(a) Table 9: Differences in past incomes for switchers.

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TABLE 10
YEAR OF FIRST POSITIVE INCOME FOR FOUNDERS OF S CORPORATIONS AND PARTNERSHIPS

Income Type	Share, First Positive Income in Year 1	2	3	Total
Business net income	53.5	17.7	7.8	78.9
Owner total income	88.0	5.3	1.9	95.3
Adjusted gross income	95.2	2.0	.7	97.8
Own wages	57.1	4.5	3.0	64.5
Spousal wages	55.2	4.9	2.8	62.9
Interest	74.7	5.9	2.9	83.5
Dividends	52.9	8.6	5.3	66.9
Capital gains	25.7	9.1	6.5	41.3

NOTE.—The table reports the share of founders who have their first positive income in year 1, 2, or 3. The sample of individuals used to compute these shares includes founders of S corporations and partnerships who started their business in the same year the business is established and have at least 8 years of tax filings with receipts or deductions.

(b) Table 10: Year of first positive income for founders.

6 Interpretive synthesis

- **Survey versus administrative data.** The paper shows that household surveys produce a different picture because they miss or compress both tails of self-employed income.
- **Typical person versus typical dollar.** Many self-employed people earn modest amounts, but a large share of dollars is earned by high-income, high-growth owners.
- **Risk versus return.** Self-employed income is more volatile and more cyclical, but average income and life-cycle growth are also much higher.
- **Entry.** Evidence on past paid income suggests positive selection into entry rather than entrepreneurship as a fallback after weak labor-market outcomes.
- **Exit.** Stable exit rates during the Great Recession imply that businesses are not abandoned immediately when income falls.
- **Reinterpretation.** The paper does not deny heterogeneity or nonpecuniary benefits; it shows that these are not the dominant explanation for aggregate entrepreneurial income in IRS data.

7 Short summary

- The paper uses IRS administrative tax data to study pass-through business owners.
- It constructs multiple definitions of self-employment and compares them with CPS survey definitions.
- Self-employed individuals are a small share of the population but a large share of total and business income.

- CPS data miss much of the right tail and understate large losses in the left tail.
- IRS panel estimates show higher and steeper life-cycle income profiles for self-employed individuals.
- The self-employed experience much larger income declines during the Great Recession.
- Exit rates do not rise sharply during the Great Recession.
- Entry rates are stable over time.
- Current entrants have higher paid-employment income before entry than future entrants.
- Most founders have positive total income early, even if business net income is not immediately positive.
- Overall, the administrative-data evidence weakens survey-based claims that entrepreneurship is mainly low-return, nonpecuniary, liquidity-constrained, or fallback employment.

8 Conclusion

8.1 Core conclusion

Bhandari, Kass, May, McGrattan, and Schulz show that the nature of entrepreneurship looks very different in population-scale tax data than in household surveys. Entrepreneurs, especially pass-through business owners with meaningful income, earn more and grow faster than paid-employed peers, while also experiencing greater income risk.

8.2 Economic intuition

Survey data emphasize the median self-employed worker and often miss high-income and loss-making business owners. IRS data reveal the full distribution. Once the tails are observed, entrepreneurship looks less like a low-paid job with compensating nonpecuniary benefits and more like a high-dispersion income process with substantial upside, cyclical risk, and positive selection into entry.

8.3 Incremental contribution in one sentence

The paper's incremental contribution is to replace survey-based views of entrepreneurship with population-scale administrative evidence on the income levels, income growth, risk, entry, and exit of pass-through business owners.

8.4 Takeaway

The main lesson is that entrepreneurship is heterogeneous and tail-driven. To understand who entrepreneurs are and what motivates entry, researchers need administrative data that observe both the very successful owners and the loss-making businesses that surveys have difficulty capturing.